

Q5

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Algorithm:-

START

1. Input N

2. IF $(N < 10)$, go to step 12.

3. Set i to N.

4. Set Sum to 0.

5. IF $i = 0$, go to step 10

6. Set digit to $i \text{ MOD } 10$

7. Add digit to sum.

8. Add i to $i \text{ DIV } 10$ and return to step 06.

9. Print Sum as the digit sum result.

10. Set N to sum and return to step 03.

11. Print N as the final single digit result.

12. END.

Pseudo code

Begin

Let N , be user input

While ($N \geq 10$), do

$i \leftarrow N$

$Sum \leftarrow 0$

While ($i > 0$), do

$digit \leftarrow i \text{ MOD } 10$

$Sum \leftarrow Sum + digit$

$i \leftarrow i \text{ DIV } 10$

END While

Print "Digit sum: ", Sum

$N \leftarrow Sum$

END While

Print "Final single digit: ", N

END